NEF ABUNDANCE: FRAMEWORK

1. Theorem and proof

Theorem. *Let X be a smooth projective threefold over $\mathbb{C}$ such that $K_X \sim 0$ and $q(X) = 0$. Let D be a nef Cartier divisor on X such that $\mathrm{n}(X, D) = 3$ and $D^3 = 0$. Then*

$$H^0(X, \mathcal{O}_X(mD)) \neq 0 \qquad \text{for some integer } m > 0.$$

Proof. Suppose for contradiction that $\kappa(D) = -\infty$, equivalently $H^0(X, \mathcal{O}_X(mD)) = 0$ for every integer $m > 0$.

Setup.

- Fix an ample Cartier divisor A and $\epsilon \in \mathbb{Q}_{>0}$. For integers $n \geq 1$, put

$$a_n = ((D + A/n)^3)^{1/3}, \qquad r_n = \max\{1, \lfloor \epsilon/a_n + 1/2 \rfloor\}, \qquad P_n = r_n(D + A/n).$$

  Then P_n is ample and rational Cartier, $r_n \to \infty$, and $P_n^3 \to \epsilon^3$. Indeed, $a_n \to 0$ and $|r_n a_n - \epsilon| \leq a_n/2$ for all sufficiently large n.

- Let $p_i : X \times X \to X$ be the projections, for $i = 1, 2$. Put

$$Z = \mathbb{P}_{X \times X}\big(p_1^* \mathcal{O}_X(D) \oplus p_2^* \mathcal{O}_X(D)\big).$$

  Let $S_1, S_2 \subset Z$ be the two summand sections and set

$$\mathbb{T} = Z \setminus (S_1 \cup S_2).$$

- For a divisor P on X, let $D_i, P_i, P_{n,i}$ denote the pullbacks of D, P, P_n by p_i, and also their pullbacks to Z. Set $\xi = c_1(\mathcal{O}_Z(1))$. For an integer $N > 0$, put

$$R(P, N) = P_1 + P_2 + N\xi, \qquad R_n = R(P_n, N).$$

  Whenever $|kR(P, N)|$ is used, $k > 0$ is an integer with kP Cartier.

For $x \in X$, let $\mathcal{I}_x$ be the ideal sheaf of the point x. Choose a birational morphism $f : V \to X$ from a smooth projective threefold such that the extended ideal

$$\mathcal{I}_x \mathcal{O}_V := \mathrm{Im}(f^* \mathcal{I}_x \to \mathcal{O}_V)$$

is invertible, and write $\mathcal{I}_x \mathcal{O}_V = \mathcal{O}_V(-E_x)$. On this fixed model, the extended ideal uniquely determines the effective Cartier divisor E_x. Let H_1, H_2 be ample Cartier divisors

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on V. All intersections involving these data are computed on V, and $E_x \cdot H_1 \cdot H_2 > 0$ since $E_x \neq 0$. For every effective Cartier divisor F on X,

$$\mathrm{mult}_x(F)\, E_x \cdot H_1 \cdot H_2 \leq f^*F \cdot H_1 \cdot H_2.$$

Indeed, a local equation of F is a local section of $\mathcal{I}_x^{\mathrm{mult}_x(F)}$ near x, so $f^*F \geq \mathrm{mult}_x(F)E_x$. The difference is effective, hence has nonnegative intersection with the ample divisors H_1, H_2.

For a Cartier divisor L on a smooth projective variety W, $|L|$ *separates s-jets at z* if

$$\forall u \in \mathcal{O}_{W,z}(L), \quad \exists \sigma \in H^0(W, \mathcal{O}_W(L)) :$$
$$\sigma_z - u \in \mathfrak{m}_z^{s+1}\mathcal{O}_{W,z}(L),$$

where $s \geq 0$ is an integer and σ_z denotes the germ at z. We take ϵ sufficiently small that

$$343\epsilon^3 < \tfrac{1}{4}, \qquad 40\epsilon < \tfrac{3}{8}.$$

We use the following three lemmas under the above setting. Their proofs are not included in this temporary framework.

Lemma 1.1. *For every integer $N > 0$ and all $n \gg 0$, a very general point $x \in X$ admits a smooth projective threefold V, a birational morphism $f : V \to X$ for which the extended ideal $\mathcal{I}_x\mathcal{O}_V$ is invertible, and ample Cartier divisors H_1, H_2 on V such that, writing $\mathcal{I}_x\mathcal{O}_V = \mathcal{O}_V(-E_x)$ as in Setup,*

$$f^*(P_n + ND) \cdot H_1 \cdot H_2 < 40\epsilon\, E_x \cdot H_1 \cdot H_2.$$

Lemma 1.2. *There exists an integer $N > 0$ such that, for every $n \gg 0$, $|kR_n|$ separates $8k$-jets at some $z \in \mathbb{T}$ for some integer $k > 0$ with kP_n Cartier.*

Lemma 1.3. *Let P be ample and rational Cartier, and $N > 0$ an integer such that*

$$343P^3 < \tfrac{1}{4}, \qquad (2P + ND) \cdot P^2 < 4P^3.$$

Suppose that some $x \in X$ admits a smooth projective threefold V, a birational morphism $f : V \to X$ for which the extended ideal $\mathcal{I}_x\mathcal{O}_V$ is invertible, and ample Cartier divisors H_1, H_2 on V. Write $\mathcal{I}_x\mathcal{O}_V = \mathcal{O}_V(-E_x)$ as in Setup and assume

$$f^*(P + ND) \cdot H_1 \cdot H_2 < \tfrac{3}{8}E_x \cdot H_1 \cdot H_2.$$

Then $|kR(P, N)|$ does not separate $8k$-jets at any $z \in \mathbb{T}$, for any integer $k > 0$ with kP Cartier.

We now complete the proof, assuming these lemmas. Choose N by Lemma 1.2 and take n sufficiently large for Lemmas 1.1 and 1.2, $r_n > N$, and $343P_n^3 < 1/4$. Since $P_n - r_nD$ is nef,

$$(2P_n + ND) \cdot P_n^2 \leq (2 + N/r_n)P_n^3 < 4P_n^3.$$

Take x, V, f, E_x, H_1, H_2 from Lemma 1.1. Then, on V,

$$f^*(P_n + ND) \cdot H_1 \cdot H_2 < 40\epsilon\, E_x \cdot H_1 \cdot H_2 < \tfrac{3}{8}E_x \cdot H_1 \cdot H_2.$$

Applying Lemma 1.3 with $P = P_n$ contradicts Lemma 1.2. Hence $H^0(X, \mathcal{O}_X(mD)) \neq 0$ for some integer $m > 0$. □

Draft status. This is a reformulation of the first-section argument with its three geometric inputs retained, rather than an independent verification of those inputs. The model $f : V \to X$, the divisors E_x, H_1, H_2, and the extended-ideal identity $\mathcal{I}_x\mathcal{O}_V = \mathcal{O}_V(-E_x)$ must be fixed before reduction of characteristic. This AI-assisted draft is for discussion and checking. No claim of novelty, priority, or authorship credit is made.